Ontology-Guided Agentic Tool and Skill Orchestration

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Motivation

Agents can perform tasks by invoking skills, tools (ie. MCP), APIs, and CLI

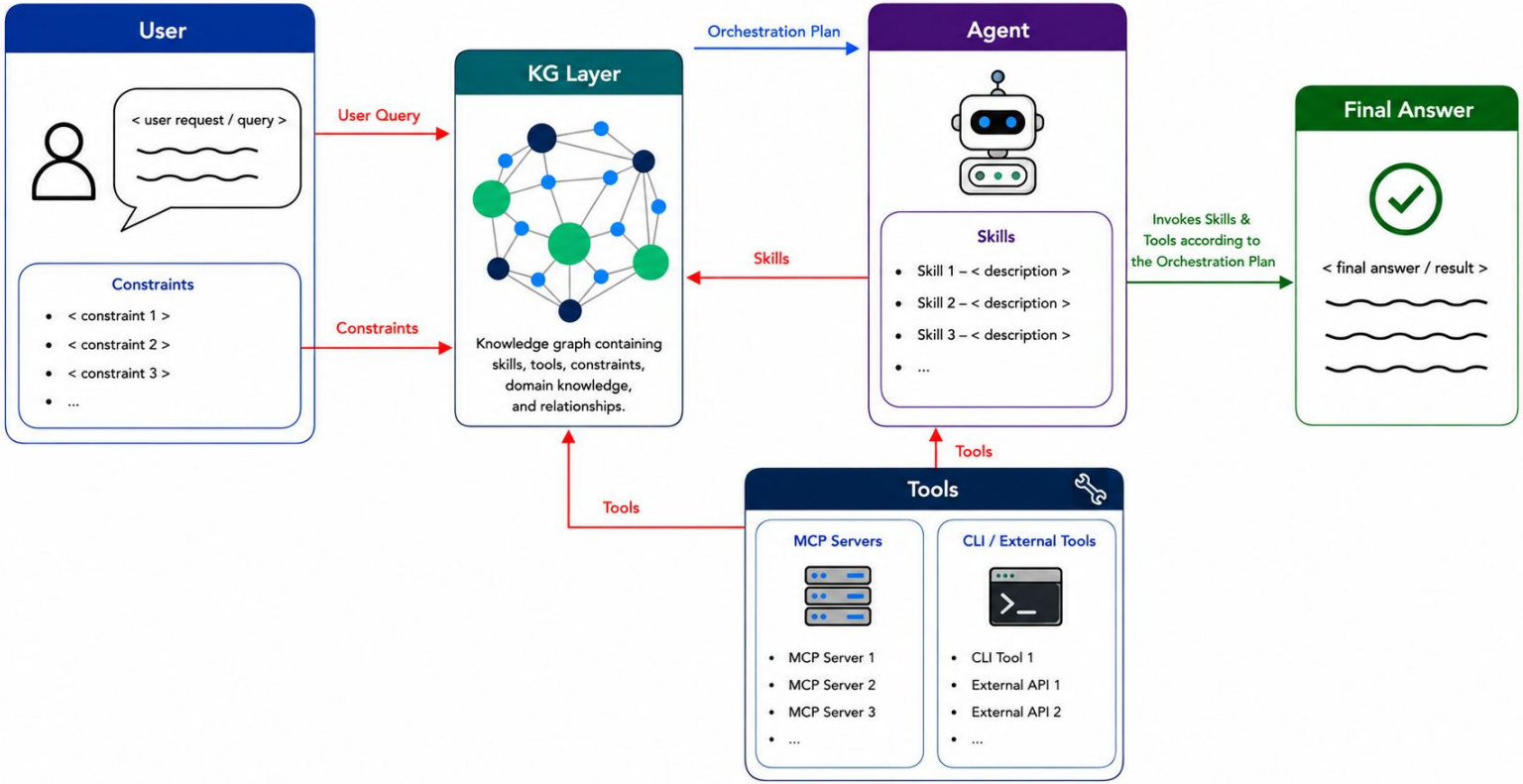
Problem shift: how can an agent successfully call a tool => which tools/skills should the agent invoke to complete a task

- Tools and skills needed to call
- In what order?
- Choosing between multiple valid orchestration plans

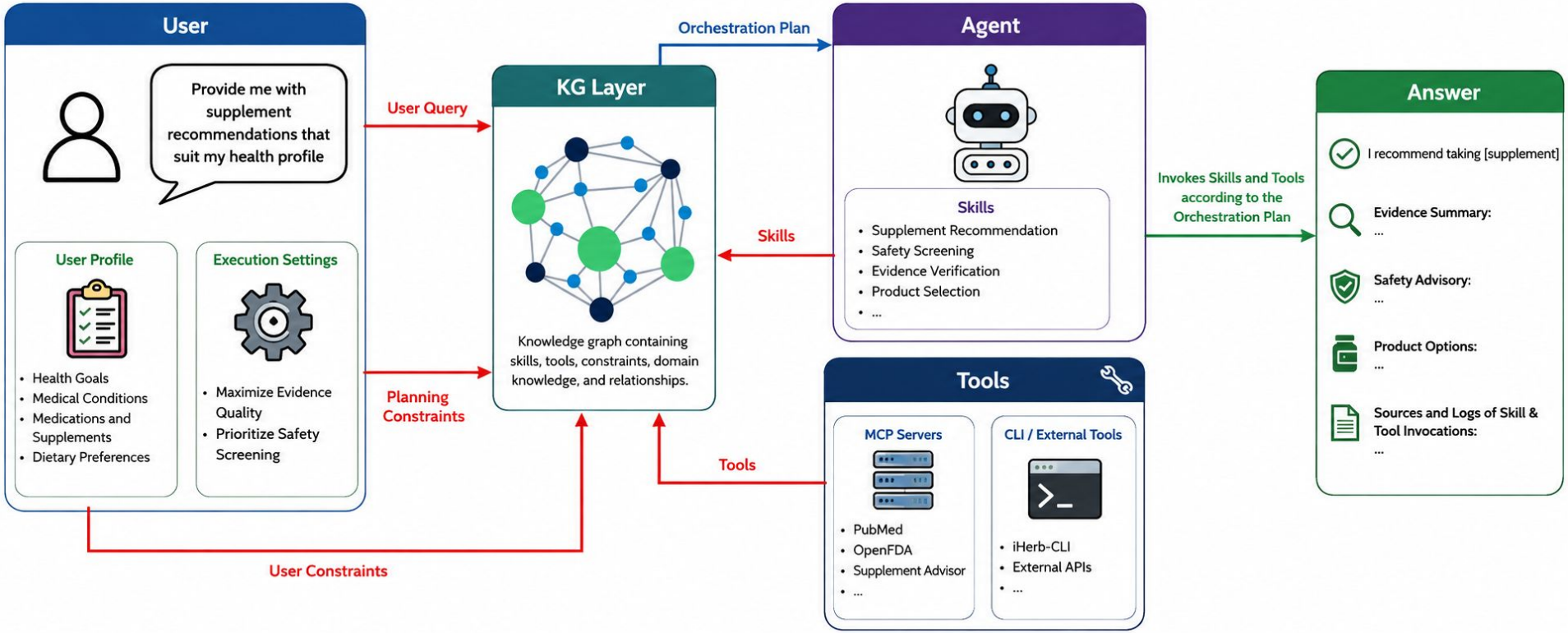
LLM-based skill and tool selection

- Rely on names and descriptions
- Incorrect/missing tool calls
- Lack of transparency and explainability for the LLM skill/tool orchestration

Concept of Operations



Personalized Supplement Recommendation



Competency Question

What orchestration plan should the agent follow to answer a user's request?

Ex. Provide me with supplement recommendations that suit my health profile.

Usage Scenario 1 (Cost-Constrained Athlete)

Ken is a **21-year-old male** college student who **exercises 5–6 days per week,** combining resistance training with cardiovascular training as he prepares to become a hybrid athlete. His primary goals are to **build lean muscle** while supporting **long-term health**, particularly **bone strength** and **cellular support**, so he can maintain an active lifestyle for years to come. He follows a **Mediterranean diet**, has **no known medical conditions or injuries**, and **does not take any prescription medications**.

Ken uses the Supplement Recommender through his LLM. Although he has a paid LLM subscription, he prefers to **minimize token usage and external tool calls** whenever possible. As a college student on a budget, he is looking for **affordable supplements** with **strong customer reviews** and **good valu**e. He wants recommendations that are **supported by scientific evidence but is willing to accept slightly less comprehensive analysis if it reduces cost**. He also wants to **avoid purchasing unnecessary or overlapping supplements that provide redundant benefits**.

Usage Scenario 2 (Safety-Critical Elderly Person)

Jessica is a woman in her **late 60s** with **osteoarthritis** who has **begun experiencing age-related declines in bone density**. She remains active through **low-impact activities** such as **walking, yoga, and tai chi** but **avoids strenuous exercise because of joint pain**. She takes **200 mg of celecoxib (Celebrex) daily** to manage her **arthritis symptoms**. Jessica's primary goal is to i**mprove her bone health** while supporting **overall longevity**, including maintaining **cardiovascular health, mobility, and healthy aging**.

She uses the Supplement Recommender through her LLM and also has a paid LLM subscription. She is not concerned about the cost of LLM usage or supplement prices. Instead, she **prioritizes recommendations that are supported by high-quality scientific evidence** and **thoroughly evaluated for safety**. She wants **high-quality supplements with minimal risk of adverse effects** and expects the system to **screen for potential contrainteractions with her prescription medication** before making any recommendations.

Usage scenario 1 (Ken) Orchestration Plan

1. **Retrieve Health Profile Skill**
2. **Retrieve Candidate Supplements Skill**
   a. **Supplement_advisor: recommend_supplement(search_query)** => candidate supplements
3. Remove redundant supplements
4. **Product Selection Skill**
   a. **iHerb-CLI: search(supplement, sort=[price])** => list of supplement products
5. Rank supplement products by cost
6. **Evidence Verification Skill**
   a. **PubMed: search_articles(search_query)** => PMIDs
   b. **PubMed: download_articles_batch(PMIDs)** => article abstracts
   c. summarize_evidence()
7. Rank supplement products by evidence
8. Select supplement product
9. Produce recommendations

Usage Scenario 2 (Jessica) Orchestration Plan

1. **Retrieve Health Profile Skill**
2. **Retrieve Candidate Supplements Skill**
   a. **Supplement_advisor: recommend_supplement()** => supplement recommendations
3. **Safety Screening Skill**
   a. **OpenFDA: search_drug_adverse_events(medication/supplement)**
   b. **OpenFDA: search_drug_labels(supplement)**
   c. **OpenFDA: search_drug_recalls(supplement)**
4. Rank by screening
5. **Evidence Verification Skill**
   a. **PubMed: search_articles(search_query)** => PMIDs
   b. **PubMed: download_articles_batch(PMIDs)** => article abstracts
   c. summarize_evidence()
6. Rank by evidence
7. **Product Selection Skill**
   a. **iHerb-CLI: search(supplement, sort=[rating])** => list of supplement products
8. Rank supplement products by rating
9. Select supplement product
10. Produce recommendations

How to solve with ontology

1. System parses the user's request and retrieves any constraints and information requirements
2. System extracts the constraints from the user's profile (ie., diagnosis, medication, health goals, etc.) and execution settings (ie.high-quality evidence, reduce supplement cost, minimize execution cost, maximize safety screening, etc.)
3. System queries the KG to identify tools and skills that can satisfy each information requirement
4. Ontology reasons over tool and skill dependencies, modeled by their expected inputs and outputs
    a. Which tools and skills are needed?
    b. What order they should execute in?
    c. Which orchestration plan best satisfies the user's constraints?
5. System creates a DAG workflow graph of the skills and tools it will execute in order
    a. Orchestration Plan and related metadata are stored in KG for provenance
6. The system selects that orchestration plan for the agent to follow